Critical Analysis and Reflection of Concept Representation in Language Models

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Abstract

Large language models (LLMs) perform strongly on many tasks, yet it remains unclear what their internal concept representations look like and how reliably they support generalization and use. This paper reviews and compares five methodological paradigms used to study concept representation in LLMs: (1) task-based probing and benchmarking via COPEN, (2) geometric analysis of categorical and hierarchical structure in representation space, (3) prompt-based evaluation of semantic relation knowledge with direct human comparison, (4) architectural interventions using concept bottlenecks for interpretable classification and controllable generation, and (5) Potemkin-style consistency evaluations that test gaps between explaining and applying a concept. For each paradigm, I summarize the core idea, evaluation setup, concept operationalization, and the main strengths and weaknesses reported in the respective work. Across these approaches, the evidence indicates that LLMs capture some categorical, relational, and hierarchical regularities, but also exhibit systematic gaps in comparison to humans and sensitivity to evaluation design and assumptions.

Keywords

large language models, concept representation, interpretability, probing methods, conceptual knowledge, human-AI comparison, methodological limitations

1 Introduction

In this paper, concept representation refers to how a language model encodes a concept so it can be expressed, generalized, and used across different contexts and tasks. Prior work defines and tests “concepts” in different ways. Some treat concepts like rule based definitions that should transfer from explanation to application, for example in Potemkin style evaluations. Others treat concepts as geometric patterns in embedding space, such as category clustering or typicality. Another line of work makes concepts explicit through architectural changes, such as concept bottlenecks. Because these choices shape what each method can actually claim, this review focuses on methodological decisions and how valid the resulting evidence is.

Studying concept representation is hard because model representations are hidden, distributed, and high dimensional, so we cannot inspect them directly. What we conclude depends heavily on how we measure them. Recent work also suggests that strong benchmark performance can hide shallow or inconsistent concept use. Models may repeat correct definitions but fail to apply them reliably, and their internal structure may not match the more coherent conceptual organization seen in humans. These issues motivate a critical review of how evidence for concepts is produced, which assumptions it relies on, and which failure modes can lead to misleading interpretations.

2 Background: Concepts and Representation in Language Models

In human cognition, concepts act as mental categories that help us organize many experiences into abstract and compressible units, while still keeping meaning. [3] describes concepts as the "glue" that binds our mental world and supports generalization across different examples. These categories often form hierarchies. For example, a "robin" is an instance of "bird", and "bird" is a subclass of "animal". This structure supports property inheritance and relational reasoning [6],[7]. Concept membership is also graded. A robin is usually seen as a more typical example of "bird" than a penguin, which reflects the typicality effect in human categorization [11]. This balance between compression and nuance is linked to stable and coherent conceptual structure in humans [10].

In artificial intelligence, early symbolic systems represented concepts explicitly using rules and ontologies [8].

Recent work reveals tensions between LLMs and human conceptual organization. Humans exhibit robust, coherent structure across estimation methods, cultures, and languages [10]. LLMs, while aligning on coarse category boundaries, often lack fine-grained nuance, show task-sensitive fragility, or prioritize compression over contextual richness [7],[11]. These discrepancies underscore the critical role of methodology: different probes may elicit superficial patterns rather than genuine conceptual knowledge, raising questions about causality, validity, and human-likeness.

3 Methodologies for Studying Concept Representation

Researchers have proposed many approaches to study how large language models (LLMs) represent concepts. In this presentation, I mainly focus on five selected methods. Each represents a distinct way of operationalizing “concept,” with different design choices, assumptions, and evaluation strategies.

3.1 COPEN Benchmark

Inspired by conceptual schemas in knowledge representation, [6] comprehensively evaluate the conceptual knowledge of pre-trained language models (PLMs) by addressing three questions: (1) Do PLMs organize entities by conceptual similarities? (2) Do PLMs know the properties of concepts? (3) Can PLMs correctly conceptualize entities in context? To investigate these questions, the paper proposes three probing tasks:

- **Conceptual Similarity Judgment (CSJ)** CSJ examines whether PLMs organize entities by conceptual similarity. Given a query entity, the model must select the most conceptually similar entity from a set of candidates. For example, although UK may frequently co-occur with Dolly, the model should select Grumpy Cat, as it shares the same superordinate concept.

- **Conceptual Property Judgment (CPJ)** CPJ evaluates whether PLMs know conceptual properties as general abstractions of factual knowledge. Given a property statement (e.g., "have feathers"), the model must judge whether it holds for a specific concept and for a concept chain, thereby testing understanding of property transitivity across hierarchical concepts.
- **Conceptualization in Contexts (CiC)** CiC assesses whether PLMs can correctly conceptualize entities within context. Given an entity mentioned in a sentence, the model must choose the most appropriate concept from a taxonomy. This requires both disambiguation and distinguishing between different levels of abstraction. For example, in "Dolly is running on the grassland," Dolly should be conceptualized as an Animal, not necessarily as a Mammal.

Also, The authors included various widely used PLMs and three probing methods in the experiment, Investigated PLMs are Masked language models (BERT, RoBERTa), Autoregressive language models (GPT-2, GPTNeo) and Sequence-to-sequence language models (BART, T5) for Probing methods, they used **Zero-shot probing:**Reformulates probing tasks into the format of pretraining language modeling objectives, enabling PLMs to perform the tasks without additional training. **Linear probing:** Adds a shallow linear classifier on top of PLM contextual representations and trains only the classifier while keeping PLM parameters fixed. **Fine-tuning:** Adapts PLMs to the tasks by training all PLM parameters on the task data with task-specific objectives.

Table 1 provides a structured comparison of the three probing tasks introduced in COPEN. The experimental results shown in Figure 1 indicate that all PLMs achieve better-than-random performance across tasks when evaluated with zero-shot or linear probing, suggesting that pretraining on large-scale text enables them to acquire a certain degree of conceptual knowledge. However, even after fine-tuning, model performance remains substantially below human levels, highlighting the need for concept-aware pretraining strategies. Furthermore, the similar accuracy observed across different pretraining objectives suggests that no current objective offers a distinct advantage for concept understanding, and that further improvements may require more targeted pretraining designs.

The authors also discuss several limitations. First, COPEN is restricted to English corpora, which limits its applicability to PLMs trained on other languages; future work aims to extend the benchmark to a multilingual setting. Second, the study does not evaluate very large models such as GPT-3 or PaLM due to limited access, instead experimenting with T5-11B, and the findings indicate that acquiring robust conceptual knowledge remains challenging for existing pretraining methods. Finally, the training and evaluation of large PLMs require substantial computational resources, resulting in significant energy consumption and environmental impact.

Model	CSJ			CPJ						CiC		
				Instance-Level			Chain-Level					
	ZP	LP	FT	ZP	LP	FT	ZP	LP	FT	ZP	LP	FT
Random	4.8	4.8	4.8	50.0	50.0	50.0	7.2	7.2	7.2	27.7	27.7	27.7
$\text{BERT}_{\text{BASE}}$	**20.3**	$\mathbf{16.1}_{0.21}$	$27.3_{0.86}$	49.4	$61.6_{0.28}$	$68.1_{0.98}$	**22.5**	$\mathbf{24.2}_{1.22}$	$\mathbf{23.2}_{1.22}$	37.6	$34.3_{0.59}$	$49.5_{0.60}$
$\text{RoBERTa}_{\text{BASE}}$	15.5	$12.0_{0.21}$	$22.3_{0.51}$	49.2	$61.9_{0.13}$	$72.0_{0.54}$	21.6	$13.1_{1.67}$	$18.3_{1.22}$	31.4	$30.0_{1.98}$	$52.6_{1.02}$
$\text{GPT-2}_{\text{BASE}}$	7.9	$4.3_{0.24}$	$20.1_{0.23}$	51.5	$64.8_{1.14}$	$70.4_{0.72}$	14.7	$14.4_{0.92}$	$20.3_{2.01}$	32.3	$34.5_{2.08}$	$\mathbf{54.2}_{0.12}$
$\text{GPT-Neo}_{\text{125M}}$	7.9	$11.0_{0.20}$	$18.3_{0.42}$	52.2	$62.2_{0.21}$	$68.2_{0.62}$	22.5	$15.0_{2.01}$	$19.0_{2.81}$	32.6	$39.6_{0.93}$	$47.4_{0.25}$
$\text{BART}_{\text{BASE}}$	14.4	$8.4_{0.10}$	$21.0_{0.50}$	48.7	$58.5_{0.27}$	$68.2_{0.86}$	20.6	$10.5_{1.22}$	$16.7_{0.80}$	33.6	$\mathbf{43.7}_{1.19}$	$51.3_{1.56}$
T5_{BASE}	15.2	$4.9_{0.21}$	$\mathbf{27.9}_{0.60}$	**55.9**	$\mathbf{66.9}_{0.25}$	$\mathbf{72.5}_{0.28}$	22.5	$18.0_{0.46}$	$18.0_{3.95}$	**42.3**	$24.7_{0.66}$	$53.2_{0.18}$
Human	79.5	79.5	79.5	91.4	91.4	91.4	91.2	91.2	91.2	85.6	85.6	85.6

Figure 1: fig:Accuracies (probing. LP: Linear probing. FT: Fine-tuning. LP and FT results are Meanstandard deviation over three random trials. Human performance is obtained by ordinary people trained with a few instances. from [6]

3.2 The Geometry of Categorical and Hierarchical Concepts in Large Language Models

This paper [4] builds upon prior findings that binary semantic contrasts defined by counterfactual word pairs (e.g., *man* → *woman*) correspond to linear directions in embedding space, and that causally independent concepts (such as gender and language) tend to appear orthogonal [5]. However, many semantic concepts, particularly categorical and hierarchical ones, cannot be captured through simple binary oppositions. For example, concepts such as "is an animal" or taxonomic hierarchies like *animal* ⊃ *mammal* ⊃ *dog* lack a clear geometric explanation under the earlier framework. The paper therefore investigates how categorical and hierarchical semantic structures are encoded in large language model (LLM) representation spaces.

The authors extend the Linear Representation Hypothesis in three main ways. First, they generalize binary concept directions into full vector representations. Second, they show that categorical concepts (e.g., {*mammal*, *bird*, *reptile*, *fish*}) can be represented geometrically as polytopes, where each vertex corresponds to the vector of a category element. Third, they demonstrate that semantic hierarchy is reflected geometrically through structured orthogonality relations between concept representations. In particular, parent-child and sibling relations within a taxonomy correspond to specific linear and angular relationships in embedding space.

To empirically validate their framework, the authors extract more than 900 noun concepts from the WordNet hierarchy and estimate their vector representations in two large models, *Gemma* and *LLaMA-3*. They then analyze geometric properties such as cosine similarity, orthogonality, and subspace relations to determine whether the learned representations align with WordNet's semantic structure. The results show that the geometric organization of concept vectors closely reflects the underlying semantic hierarchy, supporting their theoretical claims.

The approach assumes that semantic concepts can be represented linearly and that hierarchical relations can be captured through structured vector geometry. While the work provides a rigorous formal framework and strong empirical validation, it relies on idealized linear assumptions and focuses primarily on WordNet noun taxonomies. Moreover, the analysis examines static final-layer representations and does not fully address how concept encoding may vary dynamically across contexts.

3.3 Evaluation of Semantic Relation Knowledge of Pretrained Language Models and Humans

[1] propose a comprehensive evaluation framework to measure how much semantic relation knowledge pretrained language models (PLMs) acquire, and how this compares to humans. The key motivation is that earlier probing work largely focused on hypernymy alone and did not evaluate human performance on the same task, leaving an incomplete picture of what PLMs actually know about semantic relations. To address this, the authors

Aspect	Conceptual Similarity Judgment (CSJ)	Conceptual Property Judgment (CPJ)	Conceptualization in Contexts (CiC)
Main Question	Do PLMs organize entities by conceptual similarity?	Do PLMs know conceptual properties?	Can PLMs correctly conceptualize entities in context?
Task Type	Multiple-choice classification	Binary sentence classification	Multiple-choice classification
Input	Query entity + candidate entities	Property statement (e.g., "have feathers")	Sentence containing an entity mention
Output	Most conceptually similar entity	True / False judgment	Most appropriate concept from a concept chain
Conceptual Focus	Instance-of relation	Conceptual properties and transitivity	Subclass-of hierarchy and contextual disambiguation
Hierarchy Involved?	Implicit (same superordinate concept)	Yes (property transitivity across concept chains)	Yes (explicit concept chain selection)
Data Collection	Automatic collection + co-occurrence filtering	Automatic collection + human annotation	Sentence collection + human annotation
Key Finding	PLMs better distinguish coarse-grained concepts	Conceptual transitivity is difficult; conceptual hallucination occurs; word co-occurrence causes hallucination	PLMs over-rely on memory; hierarchy understanding is harder than disambiguation
Cognitive Difficulty	Similarity-based categorization	Abstract property reasoning	Context-sensitive hierarchical reasoning

Table 1: Comparison of the three probing tasks in COPEN.

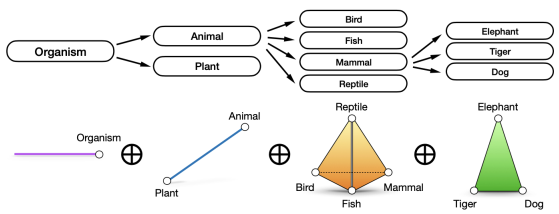

Figure 2: Illustration of hierarchical concept relations from [4]

introduce a prompt-based evaluation setup covering six relations: hypernymy, hyponymy, holonymy, meronymy, antonymy, and synonymy, and they explicitly target not only relation-specific behavior but also previously understudied meta-properties of relations.

A central contribution of the framework is its comparative design. First, the authors compare models against humans on the same probing tasks to quantify the gap relative to a "human ceiling." Second, they compare two families of models that differ in pretraining objectives—masked language models (MLMs) vs causal language models (CLMs)—and within each family they examine the impact of model size, ranging from millions to billions of parameters. This design allows them to analyze which factors (e.g., bidirectional context in MLMs) may facilitate learning semantic relations.

Methodologically, the evaluation goes beyond simple accuracy by introducing six metrics that capture different aspects of relation knowledge: soundness, completeness, symmetry, asymmetry, prototypicality, and distinguishability, where prototypicality and distinguishability are newly introduced in this work. The framework operationalizes evaluation through ranked response lists: using a prompt (probe), humans and models produce distributions over possible relata, which are converted into comparable ranked lists (with cutoffs chosen per metric). This enables fair human–model comparisons and supports analysis of both relation-level properties (e.g., prototypicality) and cross-relation confusions (distinguishability).

Empirically, the authors conduct extensive experiments (including both MLMs and CLMs) and find a significant knowledge gap between humans and models for almost all relations. Antonymy is the main exception where models perform comparatively well, but the study also reports an "antonymy bias": models often misrecognize non-antonymy relations as antonymy, suggesting systematic confusion patterns. Furthermore, MLMs consistently outperform CLMs across relations and metrics, supporting the idea that bidirectional context in MLM pretraining is important for semantic relation recognition. Finally, the results show that increasing model size does not guarantee monotonic improvement, indicating that scale alone is not sufficient for acquiring robust semantic relation knowledge.

3.4 Concept Bottleneck Large Language Models

[9] introduce Bottleneck Large Language Models (CB-LLMs), a framework for building inherently interpretable LLMs for both text classification and text generation tasks. Their contributions are threefold. First, they propose a novel framework consisting of CB-LLMs (classification) and CB-LLMs (generation), which aim to combine high predictive performance with clear interpretability. The framework is designed to match the performance of standard black-box LLMs while offering transparency that conventional LLMs typically lack.

Second, in the classification setting, CB-LLMs (classification) match the accuracy of standard black-box models and achieve a 1.5× higher average rating compared to existing methods on faithfulness evaluation. This indicates that the model provides improved interpretability without sacrificing predictive performance.

Third, in the generation setting, CB-LLMs (generation) match the performance of standard black-box generation models while enabling controllable and understandable generation. The framework allows further interaction between users and the model. In addition, the authors claimed they develop the first inherently interpretable LLM chatbot capable of detecting toxic queries and producing controllable responses.

The CB-LLM classification framework can be described in five steps:

(1) The input text is provided to the model.
(2) The model predicts human-interpretable concepts from the input.
(3) These predicted concepts form a *concept bottleneck*, forcing the downstream decision to depend explicitly on concept-level representations.
(4) A task classifier operates on the concept representations to produce the final classification output.
(5) The final prediction is interpretable because it can be explained through the intermediate concept predictions.

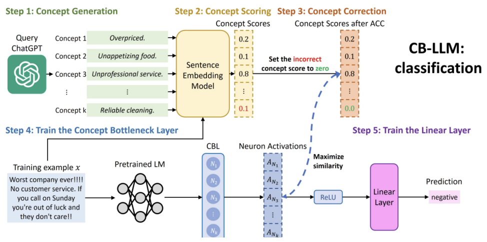

Figure 3: The overview of CB-LLMs (classification) from [9]

Experimentally, CB-LLMs (classification) achieve accuracy comparable to standard black-box models while obtaining a 1.5× higher average rating on faithfulness evaluation compared to prior approaches. In the generation setting, CB-LLMs (generation) match the performance of standard black-box generation models while providing controllable and interpretable outputs.

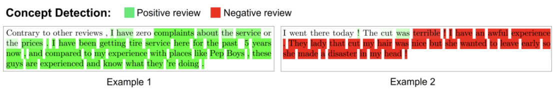

Figure 4: An example of how neurons in CB-LLMs (generation) detect the concepts. A deeper color means higher neuron activations from [9]

In the generation setting, the CB-LLM framework introduces an interpretable concept-level representation that guides text generation. This enables controllable and understandable generation while maintaining performance comparable to standard black-box LLMs. The authors further demonstrate an inherently interpretable LLM chatbot that detects toxic queries and generates controllable responses.

3.5 Potemkin Understanding in Large Language Models

[2] argues that strong benchmark performance does not necessarily imply that a large language model (LLM) *understands* the tested concepts, even when the benchmark is widely trusted for evaluating humans. The authors first introduce a framework that makes explicit the assumptions required to interpret benchmarks (e.g., AP exams, math competitions, coding challenges) as tests of understanding. Their key observation is that these benchmarks are reliable for humans because the space of *human misunderstandings* of a concept is relatively small and structured; therefore, well-designed questions can rule out most misunderstandings

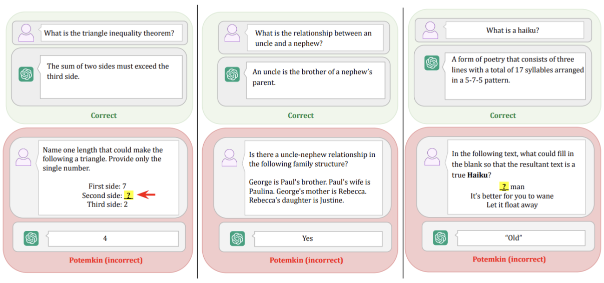

Figure 5: Examples of potemkins. In each example, GPT-4o correctly explains a concept but fails to correctly use it from [2]

(e.g., a person cannot pass a solid game-theory exam without understanding ideas such as Nash equilibria or dominated strategies). The framework implies that such benchmarks are only valid tests for LLM understanding if the space of *LLM misunderstandings* is structured similarly to the human one. If LLM misunderstandings differ, then an LLM can succeed on benchmarks without possessing the underlying conceptual competence.

When an LLM performs well on tasks that would indicate understanding in humans, yet exhibits a pattern of failures that is incompatible with any plausible human misunderstanding, the paper calls this a *potemkin*. The authors illustrate this with an example involving an ABAB rhyming scheme: an LLM can give a clear and correct explanation of ABAB rhyme, but then fails to generate text that actually rhymes in ABAB form; moreover, it may even acknowledge that its own output does not rhyme. This combination of correct and incorrect behavior is described as irreconcilable with how a human would respond, and thus constitutes a potemkin.

The paper proposes two procedures to estimate how common potemkins are in LLMs. The first targets a specific, practically important category of potemkin: the gap between an LLM's ability to *explain* a concept and its ability to *apply* it. To measure this, the authors construct a benchmark dataset spanning three domains—literary techniques, game theory, and psychological biases—designed to elicit explanation–application inconsistencies. The second procedure is domain- and structure-agnostic: it does not assume a particular form of potemkin, but as a result it only provides a *lower bound* on the prevalence of potemkins.

These methods work well together, but each has its own trade offs. Task based and geometric methods give controlled tests of what the model represents. Intervention and critical comparison methods focus more on causality and how close the model is to humans. Overall, the results suggest LLMs know some concepts, but they do not match human thinking in important ways.

To clarify how these approaches differ in their operationalization of "concept," Table 2 compares the five paradigms along three methodological dimensions: evidence type, definition of concept, and what is actually measured.

4 Critical Analysis of Methodologies

Table 3 summarizes how the five paradigms differ in terms of their concept focus, evaluation strategy, models used, and reported advantages and disadvantages. Rather than treating these differences as isolated limitations, this section briefly explains what the table reveals about the methodological landscape.

Method	Method type	Probing / evidence setup	What "concept" means in this method	What the method tests / measures
COPEN Benchmark [6]	Task-based probing benchmark	Three tasks (CSJ/CPJ/CiC) evaluated via zero-shot probing, linear probing, and fine-tuning	Concept knowledge as behavioral competence across similarity, conceptual properties (incl. transitivity), and context-based conceptualization	Task accuracy across three concept abilities; comparison of probing regimes (zero-shot vs linear vs fine-tune)
The Geometry of Categorical and Hierarchical Concepts in Large Language Models [4]	Geometric representation analysis	Estimate concept vectors from WordNet-derived concepts; analyze angles, orthogonality, subspace structure	Concepts as vectors; categories as polytopes; hierarchy encoded via structured geometric relations	Alignment between learned concept geometry and WordNet semantic hierarchy (e.g., orthogonality/subspace relations)
Evaluation of Semantic Relation Knowledge of Pretrained Language Models and Humans [1]	Prompt-based benchmarking + human ceiling	Prompt humans and models to produce relata distributions; convert to ranked lists; compute multi-metric scores	Relation knowledge as producing correct relata for a relation and avoiding confusion across relations (meta-relation behavior)	Metrics over ranked lists: soundness, completeness, symmetry/asymmetry + prototypicality and distinguishability
Concept Bottleneck Large Language Models [9]	Architectural intervention (concept bottleneck)	Introduce an explicit concept bottleneck before prediction/generation; evaluate performance and interpretability/faithfulness	Concepts as explicit intermediate variables that the model must use to reach outputs	Whether forcing a concept bottleneck yields interpretable decisions while matching black-box performance
Potemkin Understanding in LLM [2]	Benchmark-validity + consistency diagnostics	Test for explain–apply inconsistencies; use a targeted dataset (three domains) plus a general lower-bound procedure	Understanding as human-like coherence across tasks; "concept" implies consistent explanation and application patterns	Prevalence of "potemkins": benchmark success paired with non-human failure patterns (especially explain vs apply gaps)

Table 2: Comparison of five approaches by method type, probing/evidence setup, and operationalization of "concept" and what is measured.

First, the paradigms define and measure "concept" in different ways. COPEN evaluates conceptual competence through task performance (similarity, property reasoning, contextualization) using probing accuracy [6]. The semantic relation framework measures how well models produce structured relational outputs and compares them to human performance [1]. The geometric approach investigates whether internal representations align with hierarchical structures derived from WordNet [4]. Concept Bottleneck LLMs introduce explicit concept variables and evaluate performance together with interpretability and controllability [9]. Potemkin-style evaluation examines consistency between explaining and applying concepts [2].

Second, Table 3 shows that the evaluation signals differ substantially. Some methods rely primarily on behavioral accuracy (COPEN), others on structured output comparison with human ceilings (semantic relations), geometric alignment (hierarchy analysis), intervention-based faithfulness (CB-LLMs), or coherence metrics such as Potemkin rates. Because the metrics differ, results are not directly comparable across paradigms. High performance in one setting does not automatically imply strong conceptual organization in another.

Third, the table highlights different trade-offs. Probing and relation-based benchmarks are relatively easy to apply across many models but can be sensitive to dataset design and prompt structure [1, 6]. Geometric analyses provide structural insight but require access to hidden representations and depend on modeling assumptions [4]. Intervention-based methods such as CB-LLMs allow more direct control over concept usage, but they modify the architecture and therefore study a constrained system [9]. Potemkin-style tests reveal gaps between definition and use, yet focus mainly on behavioral coherence rather than internal structure [2].

Overall, Table 3 makes clear that each paradigm captures a different aspect of conceptual representation. No single method provides a complete picture. Instead, they offer complementary perspectives that need to be interpreted in relation to their specific assumptions, evaluation signals, and model access requirements.

5 Conclusion

This paper reviewed five paradigms used to study concept representation in large language models: task-based probing with COPEN [6], geometric analysis of categorical and hierarchical structure [4], prompt-based semantic relation evaluation with human comparison [1], architectural concept bottlenecks for interpretability and controllable use of concepts [9], and Potemkin-style consistency evaluations focusing on explain apply gaps [2].

Across these approaches, the reviewed studies provide evidence that LLMs capture some categorical, relational, and hierarchical regularities, while also showing systematic gaps relative to human performance and consistency depending on the evaluation setup. Overall, the findings depend strongly on how "concept" is operationalized, which signals are measured, and which models are evaluated.

Limitations

In this paper, I focus specifically on methodological approaches for studying concept representation in language models. As a

Method	Concept focus	Models used (in experiments)	Advantages	Disadvantages
COPEN Benchmark [6]	Similarity, properties, context-based conceptualization	Masked LMs: BERT-base and RoBERTa-base, Autoregressive LMs: GPT-2 and GPT-Neo, Seq2Seq LMs: BART-base and T5	Decomposes concept knowledge into three targeted abilities; compares multiple probing regimes (zero-shot/linear/fine-tune), enabling more robust behavioral interpretation	Still correlational: good performance may come from shortcuts ; benchmark scope limitations (language coverage and very-large model access)
The Geometry of Categorical and Hierarchical Concepts in Large Language Models [4]	Categorical concepts and hierarchies (WordNet)	Gemma; LLaMA-3	Provides a mechanistic, representation-level account (vectors/polytopes/orthogonality) that can be validated across many concepts; yields interpretable structural predictions	Relies on linear/geometric assumptions; focuses on static representations and WordNet-style taxonomies, which may not capture context-sensitive concept usage
Evaluation of Semantic Relation Knowledge of Pretrained Language Models and Humans [1]	Six semantic relations + meta-relations (confusability/distinguishability)	Two model families by objective (MLM vs CLM), - BERT: BERT-base, BERT-large, ALBERT: ALBERT-base, ALBERT-large, ALBERT-xlarge, ALBERT-xxlarge, RoBERTa: RoBERTa-base, RoBERTa-large, OPT: OPT-125M, 350M, 1.3B, 2.7B, 6.7B, 13B, 30B, 66B and also human participants	Adds human ceiling comparison and evaluates beyond hypernymy; introduces prototypicality and distinguishability metrics to reveal relation confusions and graded structure	Prompt/metric choices and cutoffs can influence outcomes; remains correlational (evaluates outputs rather than causal use of relation knowledge)
Concept Bottleneck Large Language Models [9]	Concept bottleneck for interpretable classification and controllable generation	Llama3-8B, GPT2, RoBERTa-base, bart-large-mnli	Causal interpretability leverage: bottleneck forces concept-based reasoning; matches black-box performance while improving faithfulness rating (classification) and enabling controllable generation	Changes architecture/system design, so conclusions do not directly describe how standard LLMs represent concepts without intervention
Potemkin Understanding in LLM [2]	Benchmark validity; explain–apply gaps; non-human inconsistency patterns	Llama-3.3, Claude-3.5, GPT-4o, GPT-o1-mini, GPT-o3-mini, Gemini-2.0, DeepSeek-V3, DeepSeek-R1, Qwen2-VL	Clarifies when human benchmarks fail to test LLM understanding; provides concrete procedures to estimate prevalence of non-human misunderstanding patterns	Targeted procedure captures a specific failure class; general procedure is conservative (lower bound) and may miss many other forms of potemkins

Table 3: Comparison of five methods by concept focus, evaluated models, and paper-specific advantages and disadvantages.

result, I do not cover other important aspects of concept representation, even though they are also relevant for understanding how LLMs represent and use concepts.

AI-use Statement

I used ChatGPT[1] for support in understanding concepts by simplifying them, clarifying wording and grammar, and producing LaTeX-safe phrasing. I also used NotebookLM[2] to read and summarize research papers. In addition, I used Grok AI[3] and ChatGPT to audit the report from an academic standpoint and to check compliance with relevant guidelines. All analyses are my own. I independently verified all citations; no sentence was accepted without my review and editing, and no sources were fabricated. I take full responsibility for the final content.

[1]https://chatgpt.com/
[2]https://notebooklm.google/
[3]https://grok.com/

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Statement of Originality

I hereby declare that I have written the thesis by myself, without contributions from any sources or aids other than those indicated. I confirm that this work has not been submitted or published elsewhere in any other form for the fulfillment of any other degree or qualification.

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